# Introduction & Conclusion Guide

Citing sources · graphs & tables · writing your conclusion

Exercise Science Project · Mr. Ward

# Today's Game Plan



## 1. Cite It

Find a source in Google Scholar, grab the APA citation, and tag facts in your Introduction.



## 2. Show It

Tables and graphs that are clean, labeled, and use percent change.



## 3. Conclude It

Plan your conclusion slides, part by part, and look back at your whole project.

# What Goes in Your Introduction

## Why it matters

Who benefits, and why you care about this experiment

## 4 key facts

The background your audience needs to understand your results

## 2 real sources

Peer-reviewed articles from Google Scholar, cited in APA

## Question + hypotheses

Research question, hypothesis (If / then / because), null hypothesis

**MURPHY SAYS** Citations live in the Introduction. Your Sources slide comes right after it.

# Why Do We Cite?



## Give credit

Someone did the work. Their name goes next to their idea.



## Let readers check

Anyone can find the same article and see if you got it right.

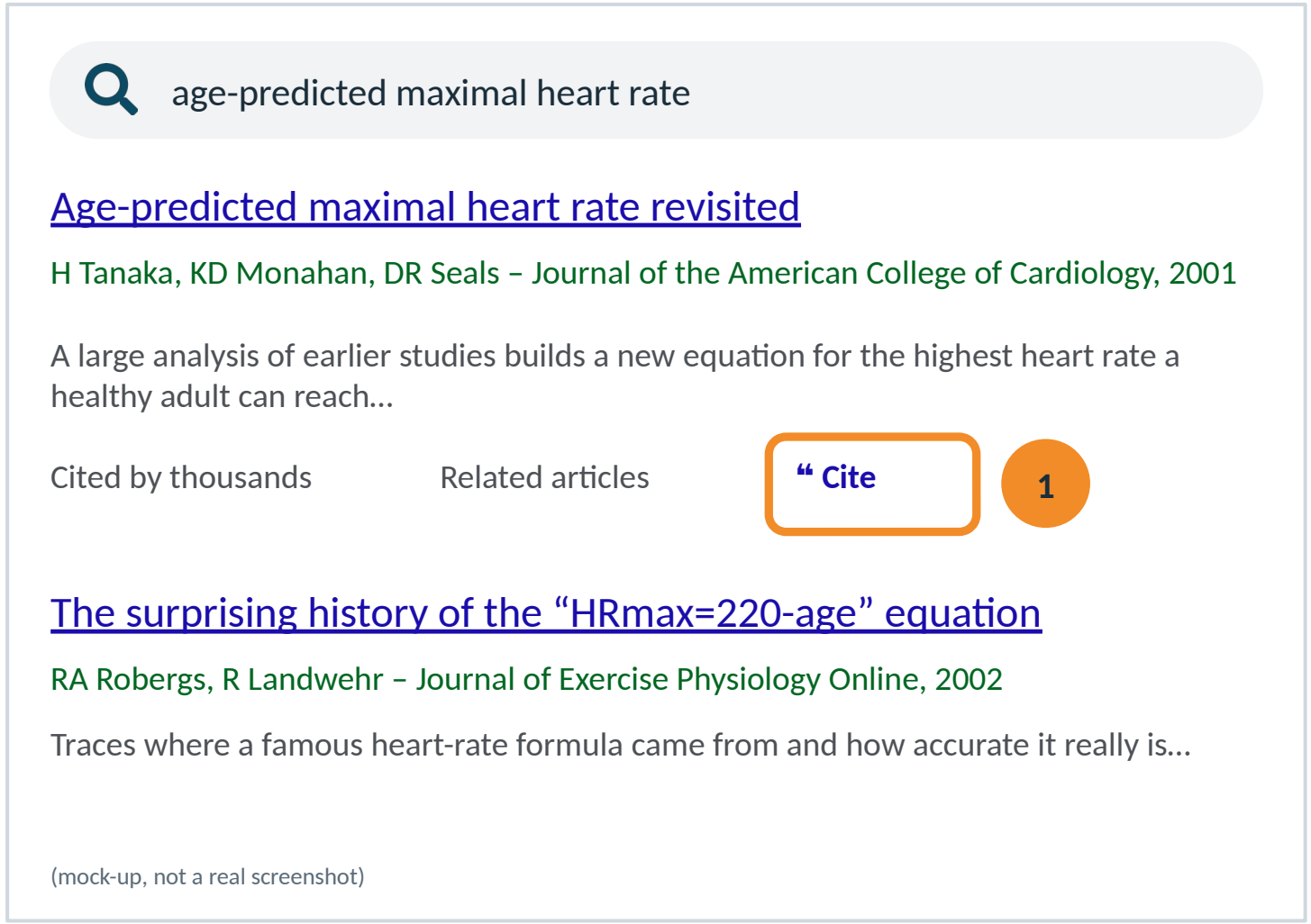


## Back up your claims

Facts from experts are stronger than “I read it somewhere.”

**Peer-reviewed article** = other scientists checked it before it was published. **Random website** = nobody checked anything.

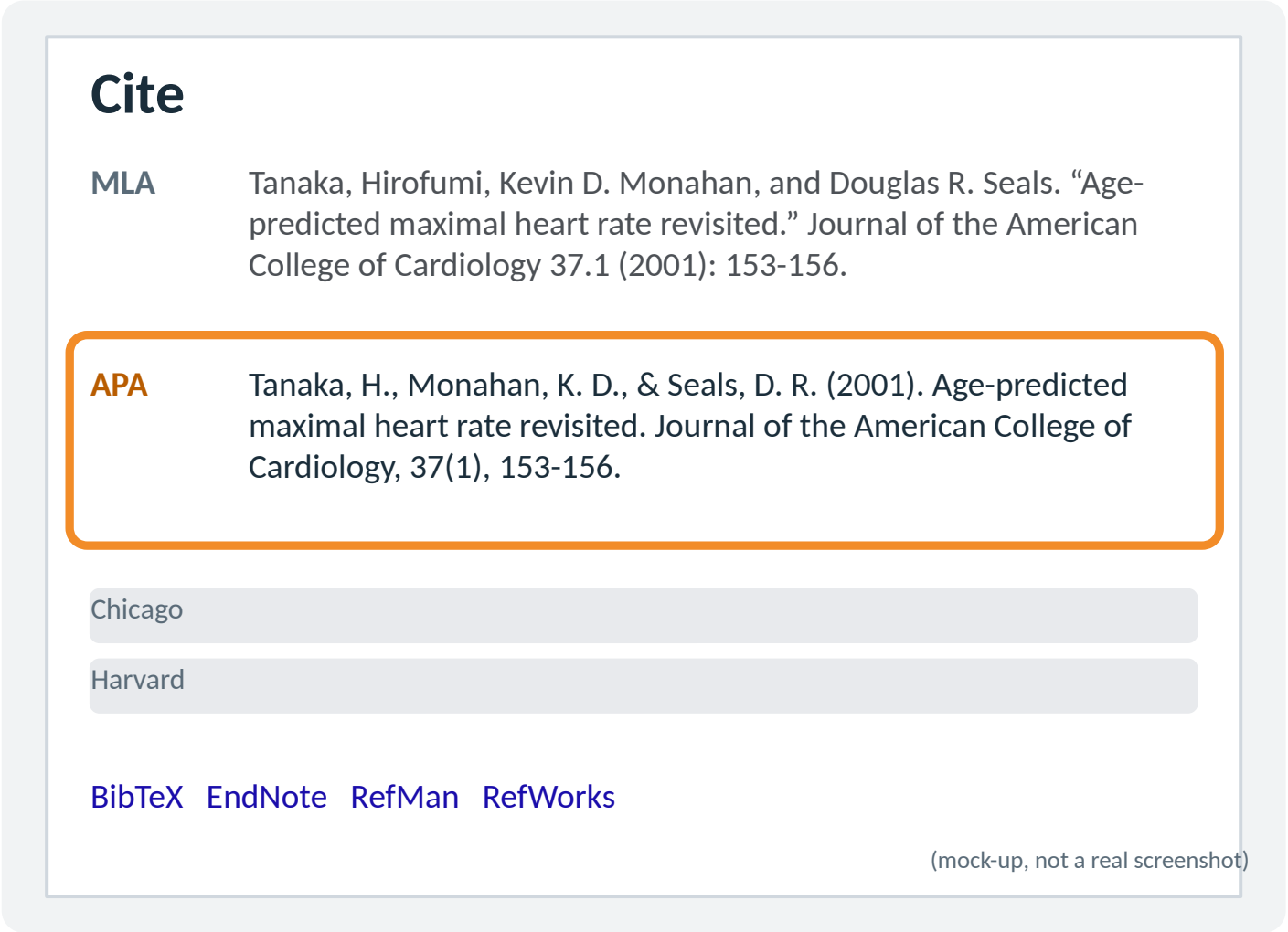
# Step 1: Find a Source in Google Scholar




- A journal article: the green line names a journal
- Authors and a year are listed
- A big “Cited by” number means other scientists trusted it
- Skip blogs and shop pages with no author or date

*Search tip: start broad (“heart rate exercise”), then narrow.*

# Step 2: Cite It in 4 Clicks




**1** **Click “ Cite under your article**

it pops up a box

**2** **Find the APA line**

ignore MLA and Chicago

**3** **Highlight it and copy**

Ctrl+C (Cmd+C on a Mac)

**4** **Paste into your Sources slide**

then do the fix-it check

# Step 3: Fix-It Check

**FROM SCHOLAR**

Tanaka, H., Monahan, K. D., & Seals, D. R. (2001). Age-predicted maximal heart rate revisited. Journal of the American College of Cardiology, 37(1), 153-156.

**AFTER YOUR FIX-IT CHECK**

Tanaka, H., Monahan, K. D., & Seals, D. R. (2001). Age-predicted maximal heart rate revisited. *Journal of the American College of Cardiology*, *37*(1), 153–156. https://doi.org/10.1016/S0735-1097(00)01054-8

- Journal name and volume in italics
- Add the DOI link if the article page shows one
- Page range uses a long dash (153–156)
- Article title: only the first word capitalized
- All authors listed, in the same order

# Citing In Text: Tag Every Fact

Scientists have estimated maximal heart rate from age for decades **(Robergs & Landwehr, 2002)**.

**Author last name(s)**

Robergs & Landwehr

**Year published**

2002

**Period goes AFTER the ")"**

(…2002).

**RULE** If you didn’t discover it yourself, it gets a tag. No tag on a fact means it looks like your own idea.

# Three Ways to Write the Tag

| Authors | At the end of a sentence | Inside a sentence |
|---|---|---|
| **1 author** | (Garcia, 2020) | Garcia (2020) reported that… |
| **2 authors** | (Robergs & Landwehr, 2002) | Robergs and Landwehr (2002) explained… |
| **3 or more** | (Tanaka et al., 2001) | Tanaka et al. (2001) found… |

**&** inside parentheses

**and** inside your sentence

*“et al.” means “and others.” Use it for 3 or more authors.*

# Use Your Own Words

**Source sentence (example):** *"During vigorous activity, the heart beats faster to deliver more oxygen to working muscles."*

### ✗ Copy, no tag

During vigorous activity, the heart beats faster to deliver more oxygen to working muscles.

**Plagiarism. Same words, no credit.**

### ✗ Copy + tag

During vigorous activity, the heart beats faster to deliver more oxygen to working muscles (Garcia, 2020).

**Still copying. Change the wording.**

### ✓ Own words + tag

When we exercise hard, our muscles need more oxygen, so the heart speeds up to supply it (Garcia, 2020).

**Your idea, their credit.**

**TRY IT** Read it, close the tab, write it from memory, then add the tag.

# Tags and References Must Match

IN YOUR SLIDES

**(Robergs & Landwehr, 2002)**

**(Tanaka et al., 2001)**

*Last name + year*

ON YOUR SOURCES SLIDE

Robergs, R. A., & Landwehr, R. (2002). The surprising history of the “HRmax=220-age” equation. *Journal of Exercise Physiology Online*, *5*(2), 1–10.

Tanaka, H., Monahan, K. D., & Seals, D. R. (2001). Age-predicted maximal heart rate revisited. *Journal of the American College of Cardiology*, *37*(1), 153–156.



Alphabetical by first author’s last name



Every tag has an entry, every entry has a tag

**REMEMBER** Two real sources, two entries. Your rubric counts them.

# Show Your Data

Tables, graphs, and percent change

# Tables: Clean and Complete

| Trial | Resting Heart Rate (bpm) | Heart Rate After Jumping Jacks (bpm) |
|---|---|---|
| 1 | 76 | 132 |
| 2 | 72 | 128 |
| 3 | 80 | 140 |
| 4 | 74 | 126 |
| … | … | … |
| **Mean** | **75.7** | **132.5** |

(sample data)

1

**Title Case headers with units**

bpm, seconds, cm

2

**All 10 trials, nothing skipped**

plus a Mean row

3

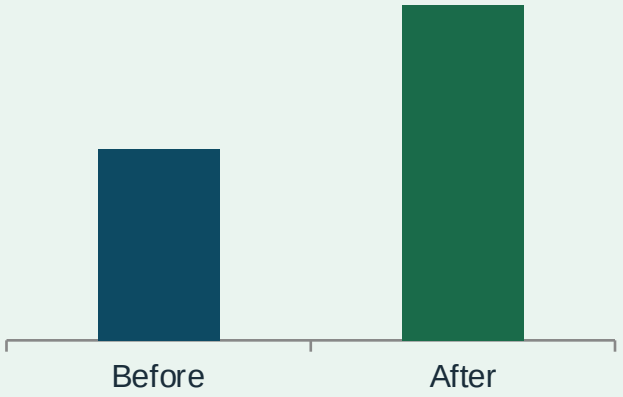
**Borders and color so it’s easy to read**

same decimals in each column

**GOOGLE SHEETS** Check the Sheets tutorial page for borders and header colors.

# Pick the Right Graph

## Bar graph




**Compare groups or averages**

Mean heart rate: before vs. after

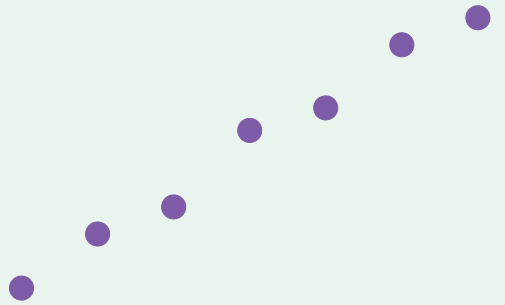
## Line graph



**Show change over time or trials**

Heart rate every 15 seconds

## Scatter plot

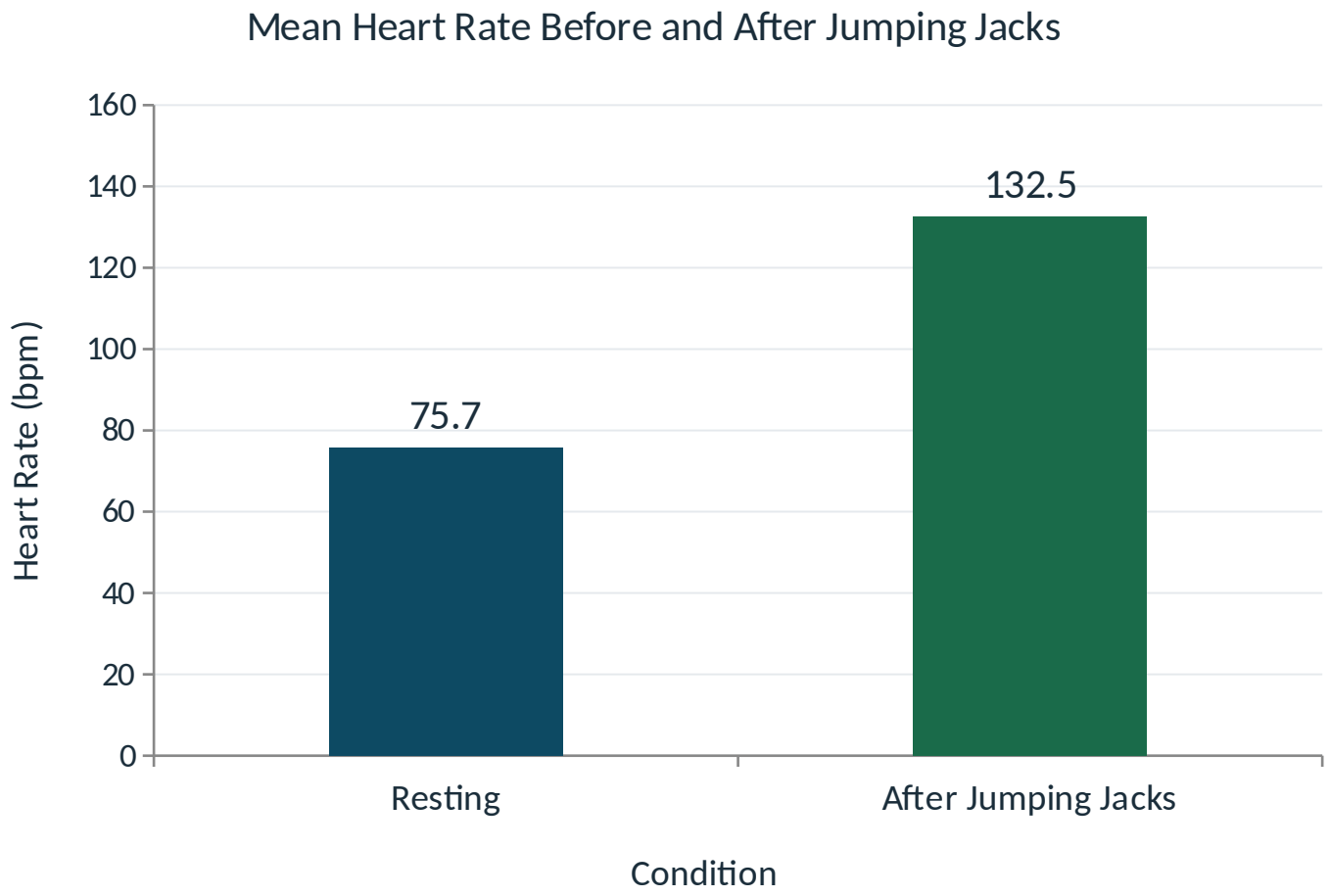


**See if two numbers are related**

Jumping jacks vs. heart rate

**NO PIE CHARTS** Pie charts show parts of a whole. Your data compares numbers.

# A Graph That Earns Full Points




1. **Title in Title Case**
   says what and when
2. **Both axes labeled with units**
   (bpm), (seconds)
3. **Y-axis starts at 0**
   no tricky zoom
4. **Values on the bars**
   easy to read
5. **Same graph type as your data**
   bar = compare averages

# Percent Change Keeps It Fair

**% Change = (After − Before) ÷ Before × 100**

## Example

Before 72 bpm, after 138 bpm

138 − 72 = 66

66 ÷ 72 = 0.917

0.917 × 100 = **91.7% increase**

## Why not just use bpm?

Person A: 60 → 100 bpm

Person B: 90 → 130 bpm

**Both rose 40 bpm. But A rose 66.7% and B rose only 44.4%.**

**MURPHY SAYS** Percent change lets you compare people who started at different heart rates.

# Plan Your Conclusion Slides

Numbers, reasons, claim, errors, and the big picture

# What Your Conclusion Slides Need

## A

### Your Numbers

*Slide: Data Analysis and P-Value*

**3 pts**

## B

### Why It Happened

*Slide: Why It Happened*

**2 pts**

## C

### Claim, Evidence, Reasoning

*Slide: Conclusion: Was Our Hypothesis Supported?*

**4 pts**

## D

### Errors and Fixes

*Slide: Errors and How We Would Fix Them*

**3 pts**

## E

### Future Expansion

*Slide: Future Expansion*

**2 pts**

**GOAL** Short phrases on the slides. Say the rest out loud. Use your real numbers.

# Part A: Your Numbers

3 pts

***What do your numbers say? Use your averages and your p-value.***

**Sentence starters**

- "Because the p-value is ___, the difference is ___, and we ___ the null hypothesis."
- Say significant (p below 0.05) or not significant.
- Say reject or fail to reject.

**Example (jumping jacks)**

The average went from 75.7 bpm to 132.5 bpm, up 56.8 bpm (75.0%). Because the p-value is below 0.001, the difference is significant, and we reject the null hypothesis.

**TIP** Put the numbers in the tiles on your slide. Keep the sentence short.

# Part B: Why It Happened

2 pts

***Why did this happen? Use science vocabulary.***

**Sentence starters**

- “During exercise, the body ___, which causes ___.”
- “Scientifically, this makes sense because ___.”

**Example (jumping jacks)**

During exercise, working muscles need more oxygen. The heart beats faster to pump more oxygenated blood to them, which causes heart rate to rise.

**TIP** Name the body parts and what they do: muscles, oxygen, blood, heart.

# Part C: Claim, Evidence, Reasoning

4 pts

## Claim

**Starter**

*Our hypothesis was ___ (supported / not supported).*

**Example**

Our hypothesis was supported. Heart rate was higher after one minute of jumping jacks.

## Evidence

**Starter**

*Our averages were ___ and ___. Our p-value was ___.*

**Example**

Our averages were 75.7 bpm and 132.5 bpm. Our p-value was below 0.001.

## Reasoning

**Starter**

*This happened because ___.*

**Example**

Working muscles need more oxygen, so the heart beats faster to deliver it.

**TIP** Claim says what you found. Evidence gives the numbers. Reasoning says why.

# Part D: Errors and Fixes

3 pts

| Error: What Could Change Our Results | Fix: What We Would Change |
|---|---|
| Because we counted the pulse for 15 seconds and multiplied by 4, one missed beat changes the result by 4 bpm. | Next time we would use a heart-rate monitor that records every beat. |
| Because counting started about 5 seconds after stopping, the heart had begun to slow. | Next time a third person would call “go” and “stop” so counting starts right away. |
| Because jumping speed changed from trial to trial, effort was not equal. | Next time we would use a metronome set to 1 jack per second. |

**Starter:** “Because ___, our results may ___. Next time we would ___.”

**TIP** Every error comes with its own fix. Name something about the design, not the people.

# Part E: Future Expansion

2 pts

***Imagine unlimited money and 5 years. How could you expand the study?***

**Sentence starters**

- "If we had millions of dollars and 5 years, we would expand this experiment by ___."
- "We would collect data from ___ and use ___ to learn ___."

**Example (jumping jacks)**

With millions of dollars and 5 years, we would track 1,000 students ages 10 to 14 with wearable sensors to compare fitness levels and recovery time.

**TIP** Be creative but realistic, and give your bigger study a clear purpose.

# Retire “Human Error”

## Don’t write

“It was human error.”

“We messed up.”

“Someone counted wrong.”

## Write the specifics

**Stopwatch started late** so trials were a few seconds short. Fix: start on a “go” call.

**Pulse counted 8 seconds after stopping** so the heart had begun to slow. Fix: count within 5 seconds.

**Jumping speed changed by trial** so effort wasn’t equal. Fix: use a metronome at 1 jack per second.

**FORMULA** What happened → how it changed the data → how you’d fix it.

# Look Back at Your Whole Project

## Your Experiment

**Part G**

- What was the strongest part? The weakest?
- What is the first thing you would redo?
- Did your results surprise you? What would you test next?

## The Whole Trimester

**Part H**

- Which skill grew the most?
- How did you and your partner split the work?
- What could you use in another class or in real life?
- What advice would you give next year's 7th graders?

**MURPHY SAYS** These are for you. Share one idea out loud when you answer questions.

# Before You Turn It In

- Part A: numbers and p-value filled in
- Part B: science words explain why
- Part C: claim, evidence, reasoning match
- Part D: errors are about the design, each with a fix
- Part E: a creative, realistic expansion
- Parts F to H: practice and reflection done